Exam style problems – set no.1

I. Continous random variable.

Task 1. The cost of producing a unit of a certain product is a continuous random variable with density

$$f(x) = \begin{cases} \frac{4}{81}x^3 & dla \quad x \in (0,3) \\ 0 & dla \quad x \notin (0,3) \end{cases}.$$

Calculate:

a) the distribution function of the random variable,

b) the probability that the cost of producing a unit of this product will not exceed 1 PLN,

c) the probability that the cost of producing a unit of this product will exceed 2 PLN.

Task 2. The random variable X has function of probability density given as

$$f(X) = \begin{cases} \frac{1}{2}\sin x & dla \quad x \in (0;\pi) \\ 0 & dla \quad x \notin (0;\pi) \end{cases}$$

a) Find the distribution function of the random variable X
b) Calculate the probability

$$P\left(\frac{\pi}{4} < X < \frac{5}{4}\pi\right)$$

Task 3. The random variable X has function of probability density given as

$$f(x) = \begin{cases} 0 & dla \quad x \leqslant 0 \\ cx & dla \quad 0 < x < 1 \\ cx^2 & dla \quad 1 \leqslant x < 2 \\ 0 & dla \quad x \geqslant 2 \end{cases}$$

Determine:

a) the constant c,

b) the cumulative distribution function

c) and calculate P (1/2 <= X <= 3/2)